

Abstract No.

Glioblastoma cell-derived APLN acts as a chemoattractant for tumor-associated myeloid cells

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Glioblastoma (GBM) progression is driven by tumor-induced immunosuppression and the massive accumulation of tumor-associated myeloid cells (TAMs). Consequently, signaling pathways governing myeloid cell tropism represent promising therapeutic targets. We previously showed that the angiogenic factor APLN signals through its receptor APLNR to regulate neovascularization and GBM invasion; additionally, TCGA database analysis revealed that APLNR expression correlates with lymphocyte-mediated immune responses.

Here, we investigated the role of the APLN-APLNR axis in GBM-immune cell interactions using primary GBM cells, murine myeloid cells (BV2 and J774), and APLN-knockout mice. Myeloid migration was assessed via Boyden chamber and μ -slide chemotaxis assays.

In human GBM xenografts, APLN expression directly correlated with TAM density in both the tumor core and invasive front. We confirmed APLN expression in primary GBM cells and APLNR expression in Iba1-positive TAMs, primary microglia and myeloid cultures. In vitro, GBM-conditioned medium stimulated myeloid cell migration, which was fully abrogated when using APLN-knockdown (AKD) GBM medium. Notably, J774 macrophages only migrated toward APLN after prolonged exposure to GBM-secreted factors, which dynamically upregulated APLNR expression.

Our findings identify tumor-derived APLN as a novel myeloid cell chemoattractant that shapes TAM distribution within the GBM microenvironment. Targeting the APLN-APLNR axis represents a promising multi-modal therapeutic strategy to simultaneously inhibit angiogenesis, tumor cell invasion, and immunosuppressive myeloid cell recruitment.



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